Table of Contents

Decision Log

Field	Value
Document ID	NOVA-DL-001
Version	0.3
Status	Draft
Owner	Laxmi Poudel
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1. Purpose

This log records project and technical decisions that do not warrant a full Architecture Decision Record. Architecturally significant decisions are recorded as ADRs in `adr/` using MADR 4.0 and are indexed in section 6 below.

A decision log that is not maintained is worse than none, because it misrepresents the current state of the project.

2. Status values

Status	Meaning

Status	Meaning
`Accepted`	Decided. Change requires a superseding entry
`Provisional`	Decided on limited evidence. Confirmation pending
`Open`	Awaiting measurement, with a stated trigger
`Superseded`	Replaced. Retained for history

3. Product and scope decisions

ID	Decision	Status	Rejected alternatives	Rationale	Accepted cost	Revisit trigger
DL-001	Primary workflow is requirement to structured test plan	Accepted	Defect report structuring; diagnostic plan generation	The only candidate whose output is objectively gradeable, which the central claim requires. Feasible within the inference constraint. Dense feedback signal from ordinary use	The problem space is crowded, so differentiation must come from the feedback loop rather than the generation	The model spike demonstrating that schema conformance is unattainable
DL-002	Local-first inference with a pluggable hosted provider	Accepted	Local only; hosted primary	Confidentiality position, zero marginal cost, and a materially harder engineering problem. The abstraction preserves a demonstration fallback	A lower output quality ceiling	Hardware proving inadequate
DL-003	Local deployment only, container composition	Accepted	Hosted demonstration instance	A recorded demonstration substitutes at lower cost	No publicly reachable instance	A publicly reachable instance becoming

ID	Decision	Status	Rejected alternatives	Rationale	Accepted cost	Revisit trigger
	as the artifact					worthwhile
DL-004	Synthetic and open-source data only	Accepted	Authentic work artifacts; mixed	The repository becomes publishable, and privacy controls become designed and tested rather than compliance-bound	Less realistic inputs	Authentic data entering the system, at which point PII detection becomes a requirement
DL-005	Single user with a multi-tenant-capable schema, authentication deferred	Accepted	Full authentication; no tenancy model	Maximum isolation-testing value per hour invested. Authentication changes the resolution point, not the data model	No authentication capability to demonstrate	Post-release
DL-006	Model fine-tuning deferred rather than excluded	Accepted	Permanent exclusion; inclusion in the initial release	The initial release produces the labelled dataset and the baseline that a fine-tune requires. Training first would be unmeasurable	Requires present-tense discipline in all material until a trained model exists	Sufficient dataset volume and a measured baseline both present
DL-007	Effort weighted unevenly across capability	Accepted	Even distribution	Even distribution yields uniformly shallow work in	The user interface will not be visually distinguished	Narrowing to a single capability area

ID	Decision	Status	Rejected alternatives	Rationale	Accepted cost	Revisit trigger
	areas			every area		
DL-008	Four playbooks	Accepted	Two; three	Sufficient spread for selection to visibly matter, and it stratifies the evaluation dataset naturally	Four prompt fragments and four dataset strata to maintain	Effort pressure triggering the scope reduction order
DL-009	Scope reduction order committed in advance	Accepted	Deciding under pressure	Decisions taken calmly are better than decisions taken in the tenth week	A capability may be removed that is later regretted	None

4. Technical decisions

ID	Decision	Status	Rejected alternatives	Rationale	Accepted cost	Revisit trigger
DL-010	Three-operation inference gateway interface	Accepted	A general inference abstraction layer	Structured generation, text generation, and embedding cover every requirement. Anything wider is speculative	Provider-specific capabilities are unreachable	A concrete requirement arising
DL-011	Client-rendered single-page application	Accepted	Server-side rendering framework; server-rendered templates with progressive enhancement	Server-side rendering yields nothing for a local single-user application. Query caching addresses the one non-trivial client concern, which is polling a running task	No server-side rendering	Hosted deployment

ID	Decision	Status	Rejected alternatives	Rationale	Accepted cost	Revisit trigger
DL-012	Outcome scores derived from the event log, never authoritative	Accepted	Storing scores as the record of truth	Recoverable following poisoning, corruption, or a change to the weighting scheme	Recomputation cost	None
DL-013	Bounded exploration over a fixed action set	Accepted	Pure exploitation; upper-confidence-bound or Thompson sampling; static mapping	The simplest mechanism that is real, scoreable, and explainable	A modest mechanism. It is not reinforcement learning and is not described as such	The action set growing materially
DL-014	Content-free logging by default	Accepted	Logging prompts for diagnostic value	The confidentiality position cannot be undermined through logs	More difficult diagnosis, offset by an explicit debug mode	None
DL-015	RFC 9457 problem-detail error representation	Accepted	Ad hoc error shapes	One representation for every failure, and one rendering component	Slightly verbose	None
DL-016	Cursor-based pagination	Accepted	Offset pagination	Correct behaviour on append-heavy relations	Opaque cursors	None
DL-017	Deterministic checks first, model-based review confined to the residual	Accepted	Model-based evaluation as the primary mechanism	Avoids making every metric dependent on a component whose reliability must itself be established	Misses nuance that a model-based judge would detect	Deterministic checks proving insufficient
DL-	Correction	Accepted	Acceptance	Directly falsifies	Sensitive to a	The

ID	Decision	Status	Rejected alternatives	Rationale	Accepted cost	Revisit trigger
018	Recurrence Rate as the primary metric		rate; evaluation pass rate	the central claim and requires no model-based judge	threshold choice, which must be stated wherever the metric is quoted	threshold proving indefensible
DL-019	Fake inference provider for integration testing	Accepted	Real models throughout	Integration tests must be fast and deterministic. Model behaviour belongs to the evaluation sub-process	Integration tests cannot detect model-specific defects	None
DL-020	Evaluation thresholds derived from measured variance	Accepted	Thresholds set from intent	An unstable gate is disabled in practice, and a disabled gate is worse than none	Thresholds cannot be set until the harness exists and variance is measured	None
DL-030	The inference runtime stays bound to loopback; containers reach it through Docker Desktop's host proxy	Accepted	Widening `OLLAMA_HOST` to `0.0.0.0` with the port firewalled to the Docker subnet; containerizing inference with GPU passthrough	Measured working on the default binding, so the alternative buys nothing and costs an unauthenticated inference API behind one firewall rule	The deployment is Docker Desktop-specific. A Linux-native engine has no host proxy and needs the widened binding	Deploying on a container runtime without a host proxy

5. Decisions pending measurement

ID	Decision	Status	Evidence to date	Next action
DL-021	Hardware provides 6 GB VRAM, not 8 GB or greater	Accepted, corrected by measurement	RTX 4050 Laptop, 6141 MiB, measured 2026-09-07	Planning had recorded 8 GB or greater. The candidate model list was revised throughout the document set

ID	Decision	Status	Evidence to date	Next action
DL-022	Model class fixed at 4B parameters rather than 8 to 9B	Accepted	Five candidates over twenty requirements. The 4B class is 2.9 to 3.9 GB and fully GPU-offloaded; both 8B candidates need 6.6 GB, run about a third on the CPU, and are 2.4 times slower for no quality gain	Settled by the full run on 2026-09-08
DL-023	Specific model and quantization	Accepted	`gemma3:4b` at its default quantization. Leads content quality at 0.988 AC coverage and 0.988 required case types over twenty requirements, no hallucinated criterion references, smallest footprint at 2.9 GB	Chosen over a 7 s median latency advantage elsewhere, because the faster candidates each carry a coverage or accuracy deficit
DL-024	Single-stage constrained generation rather than reason-then-structure	Accepted	100 of 100 valid on first attempt at single stage, across five models and twenty requirements	The complete run showed no structural weakness to justify a second inference call
DL-025	Thinking disabled on hybrid reasoning models	Accepted	Required for one candidate family. Reasoning tokens dominate generation duration and interact poorly with constrained decoding	A configuration requirement, not an optimization
DL-026	Embedding model and vector dimension	Accepted	`embeddinggemma` at 768 dimensions. Five candidates measured against 71 hand-labelled pairs: first on every discriminating metric, 77 ms warm, 681 MB resident, co-resident with the generation model inside 6 GB	Decided in ADR-0005 on the evidence in NOVA-SPK-002. Revisit when authentic corrections exist in volume
DL-027	Whether the 8B class fits within the VRAM budget	Accepted	It does not. Both candidates load at 6.6 GB against 6141 MiB, running 36% and 38% on the CPU, at 40 to 44 s median against 17.3 s	R-08 closed on the measurement. Revisit only on different hardware
DL-	Duplicate	Open	None	Derive from measured

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028	similarity and structural similarity thresholds			distributions, then fix
DL-029	Always-apply memories are held outside the vector index rather than retrieved by similarity	Provisional	A rule labelled relevant to all twenty requirements was ranked 6th to 8th by every one of five candidates, for every requirement. It accounts for the entire gap between recall@5 of 0.701 and scoped recall of 0.967	A content-free rule gives semantic ranking nothing to match on. Settle the memory entity's treatment of scope before the M3 schema

6. Architecture Decision Records

ADR	Decision	Status
0001	PostgreSQL with pgvector as the sole datastore	Accepted
0002	Job table and polling worker rather than a message broker	Accepted
0003	No agent framework	Accepted
0004	Mandatory human confirmation for memory writes	Accepted
0005	Embedding model and vector dimension	Proposed

7. Traceability

7.1 User need to verification

Need	Requirements	Components	Verification
UN-1 Retain conventions across sessions	FR-5, FR-6, FR-4	Feedback processor, memory store, retriever	Correction Recurrence Rate; extraction precision measurement
UN-2 Output in the team's format	FR-2, FR-4	Generator	Structured output validity; schema conformance tests
UN-3 Determine whether it is improving	FR-9, FR-10	Evaluation harness	Correction Recurrence Rate; evaluation pass rate; regression rate
UN-4 Understand why an output was produced	FR-8	Orchestrator, observability	Trace completeness; trace correlation tests
UN-5 Complete negative and authorization	FR-2, FR-3	Selector, generator, reviewer	Acceptance criteria coverage; required case type presence

Need	Requirements	Components	Verification
coverage			
UN-6 Avoid third-party transmission	FR-11	Inference gateway	Provider contract suite; explicit switch test
UN-7 Correct or remove a poor rule	FR-7, FR-6.5	Memory store, feedback processor	Deletion completeness; conflict detection tests

7.2 Threat to verification

Threat	Mitigation	Verification
T3 Persistent injection through memory	Validation screen, human confirmation, absence of tool surface	Adversarial scenarios targeting extraction and confirmation
T10 Cross-tenant disclosure	Data-layer scoping, server-side tenant resolution	Isolation suite as an unconditional gate
T5 Feedback poisoning	Bounded updates, clamping, minimum observations, derived scores	Poisoning scenarios; score recomputation test
T19 Elevation through injected instruction	No tool surface	Architecture inspection; adversarial suite

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft
0.2	2026-09-08	Laxmi Poudel	DL-026 accepted. DL-029 added: always-apply memories held outside the vector index
0.3	2026-09-08	Laxmi Poudel	DL-022, DL-023, DL-024 and DL-027 accepted on the full model run
0.4	2026-09-09	Laxmi Poudel	DL-030 added and accepted on the evidence in NOVA-SPK-003